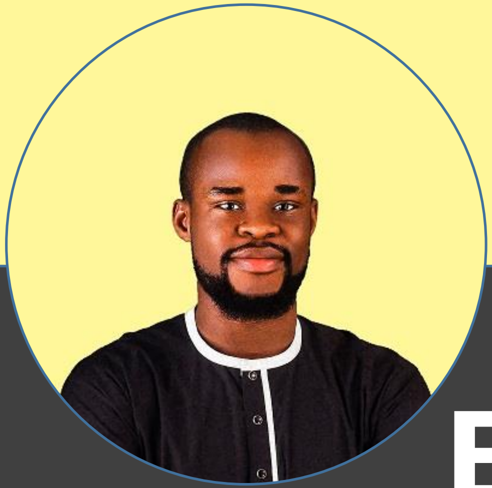


Business Intelligence Analytics

As the company prepared to celebrate its anniversary, management wanted to evaluate overall business performance and identify loyal customers for recognition. They were also concerned about increasing product return rates and needed to understand the factors driving those returns so they could make informed business decisions.

My Process

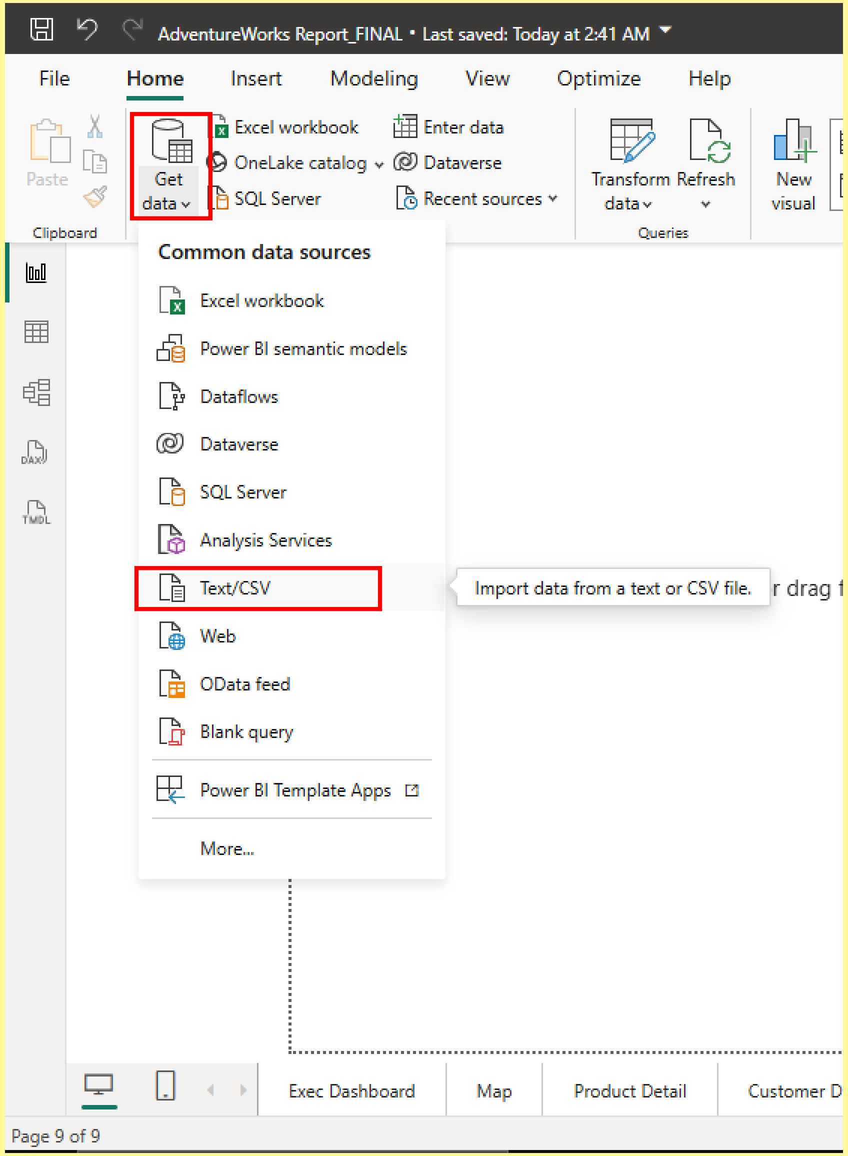
Overview

This project involved developing an interactive Power BI dashboard to analyze sales performance and key business KPIs. By consolidating revenue, profit, orders, customer, product, and regional data into a single reporting solution, the dashboard enables stakeholders to monitor performance, identify trends, and make informed, data-driven decisions.

Business Problems/Questions(EDA)

1. What is the business perfoemance based on the KPI (Revenue, Profit, Orders, return Rate)?
2. Which Customer has invesetd the most over the years?
3. Which region has produced the most revenue?
4. What products are the most sold and the most returned?
5. What category of products has the most orders?
6. Predict revenue with a 10% increase in price.
7. How are each product performing?
8. What are the top 10 products by orders?
9. What is the revenue trend over the years?
10. What is the revenue, profit and returns performances compared with previous months
11. What is the average revenue generated per customer?
12. What is the total order by the top 3 Income level and occupation?
13. Why is their an increase in return rate?

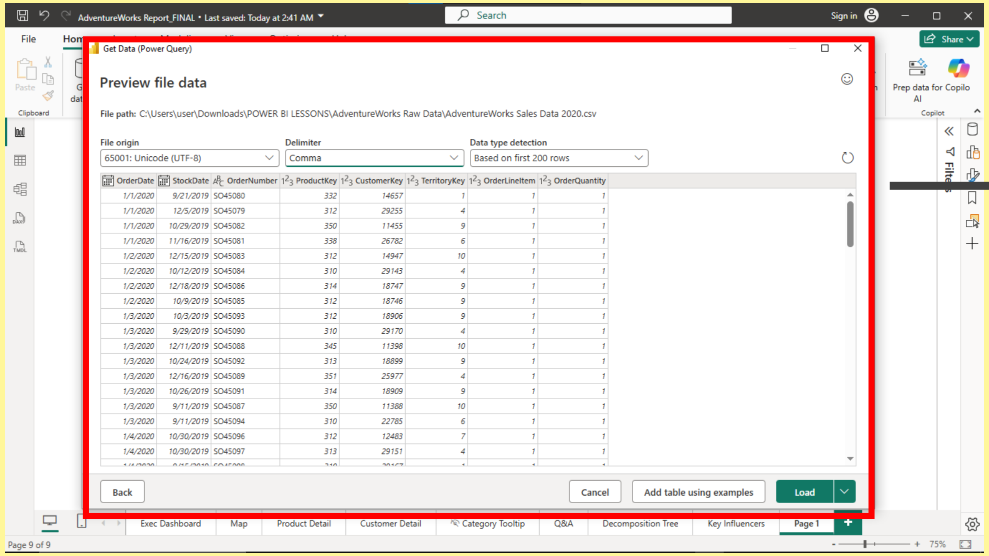
1. Importing the Files



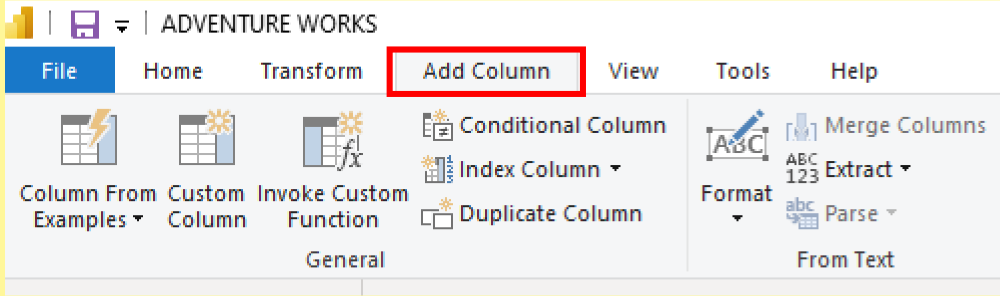
There are a number of ways to get your data into the work space incluing from databases, but for the sake of the demostrationwe will be importing Text/CSV files already ectracted from the data base using SQL.

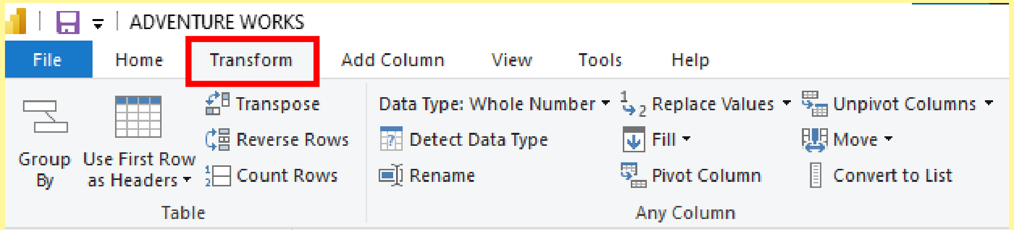
Clicking on the get data icon brings a pop up that prompts you to choose importstyle and then locate the file on your computer

2. Data Cleaning (Power Query)



After selecting your data cleaning is important is this happens through the Power Query, a powerful tool that allows irregular, missing values to be tackeled and also allows the creation of calculated columes.



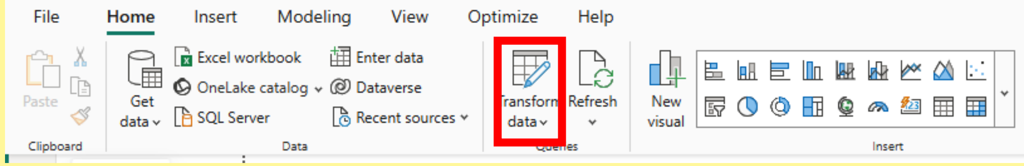


The add column and transform tabs and parts of the power query tool that allows you modify your data before creating relationships. Following the need of the business there is usually a clear idea on tables that need to be included

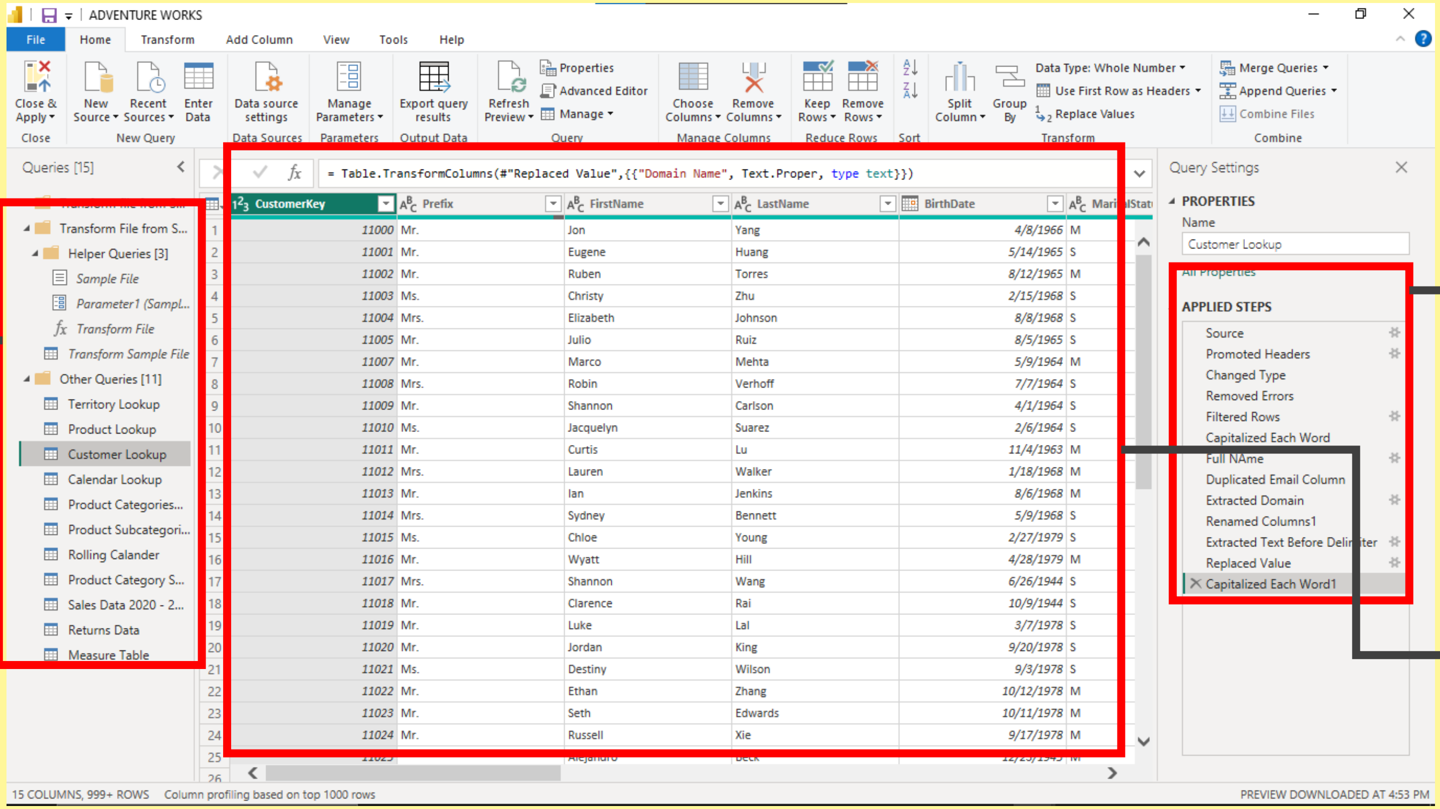
N/B: these are great tools but it is important not to overuse them cause they can make the file really heavy creating a lot new and redundant data. DAX measures are a preferred way. So calculated columes should be used only when necessary

3. Previewing Data Tables after cleaning

After cleaning the data its important to carefully observe and walk through in case anything was missed. The Transform data tab allows you access the power query



This tab shoes all the tables loaded into the power query



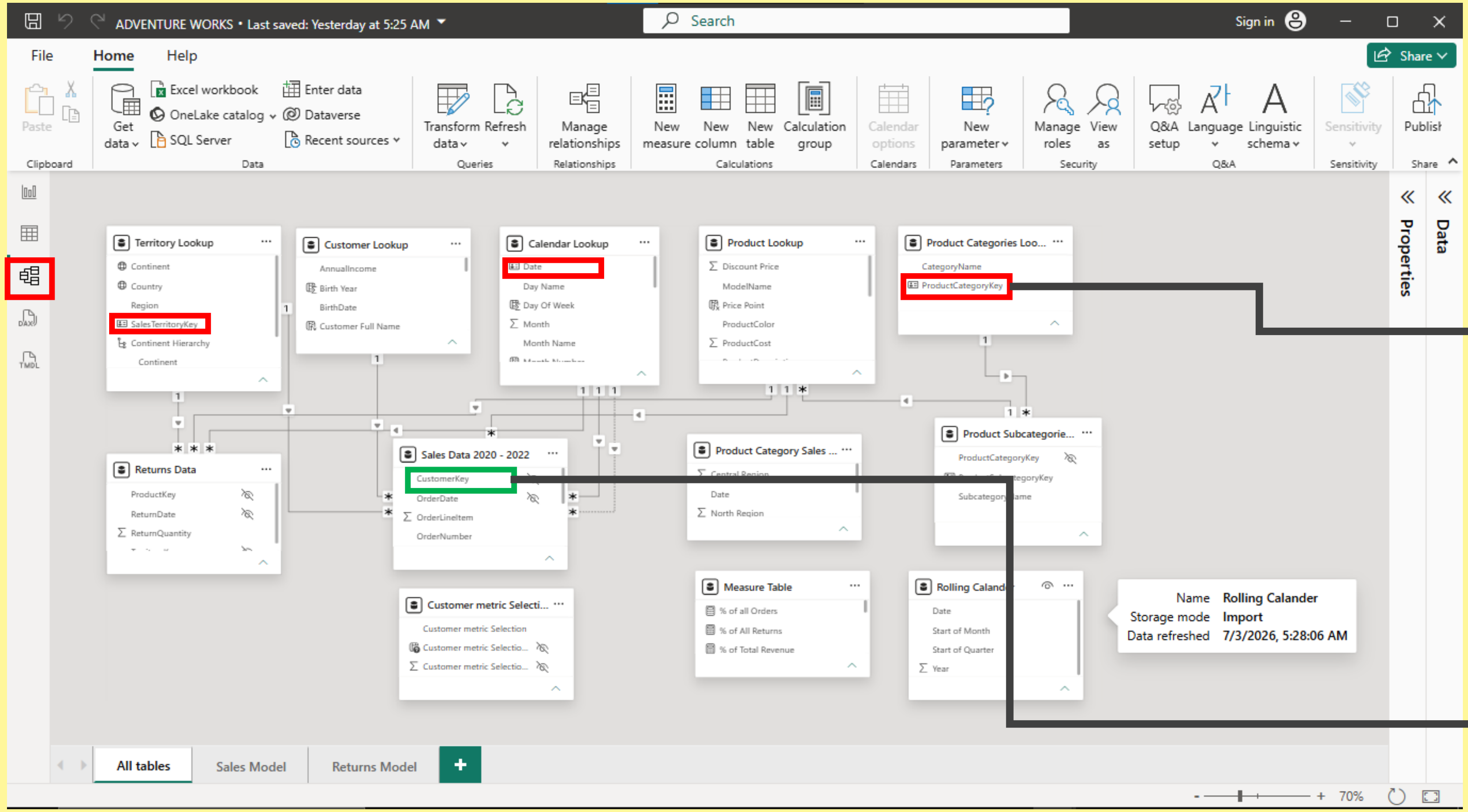
The applied steps tab showsall steps taken in transforming the data within the power query

This tab displays the current selected table

4. Data Modeling

This is a vital part of the analysis it involves creating relationshsips between previously independent tables allowing them to communicate without merging or combining them. This relationships are created through **Primary and Foreign keys.**

This tab opens the data modelling view



Columns in red are Primary keys which are unique identifiers of a table

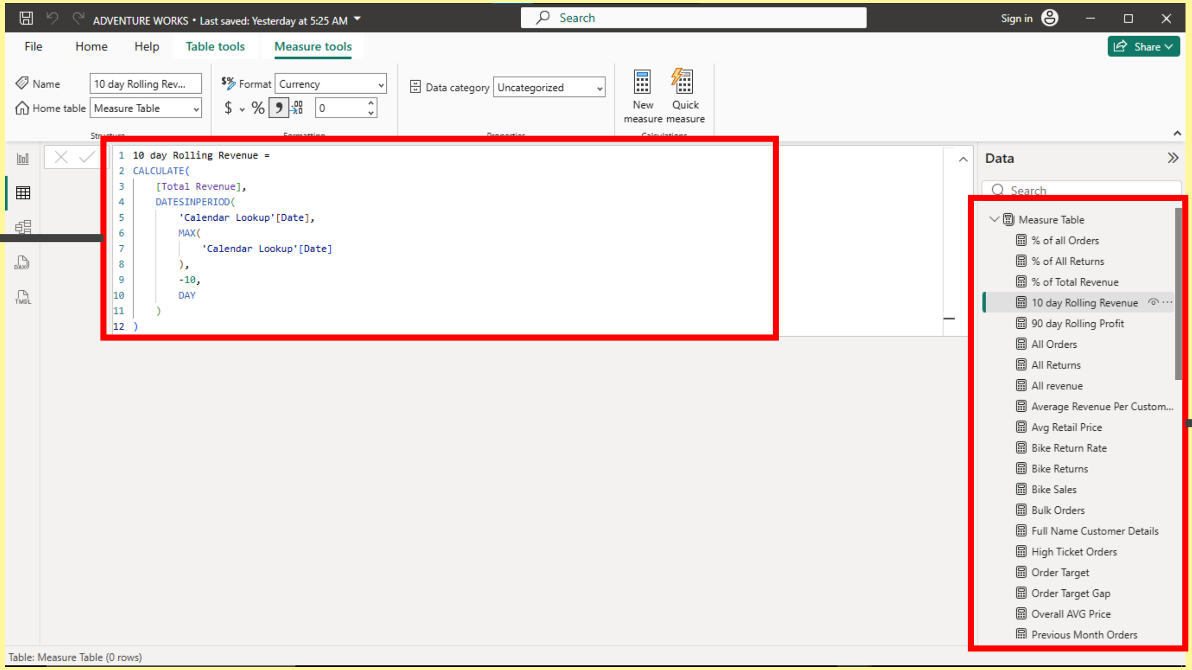
This is a foreign key it is a key that doesn't have unique entities

N/B: The pattern for connect a primary key to a foreign key is "one to many"

5. DAX

DAX measures are one of the amazing tools that allows a lot of computations and calculations without the need to create redundant calculated columns. They are a big part of the analysis process.

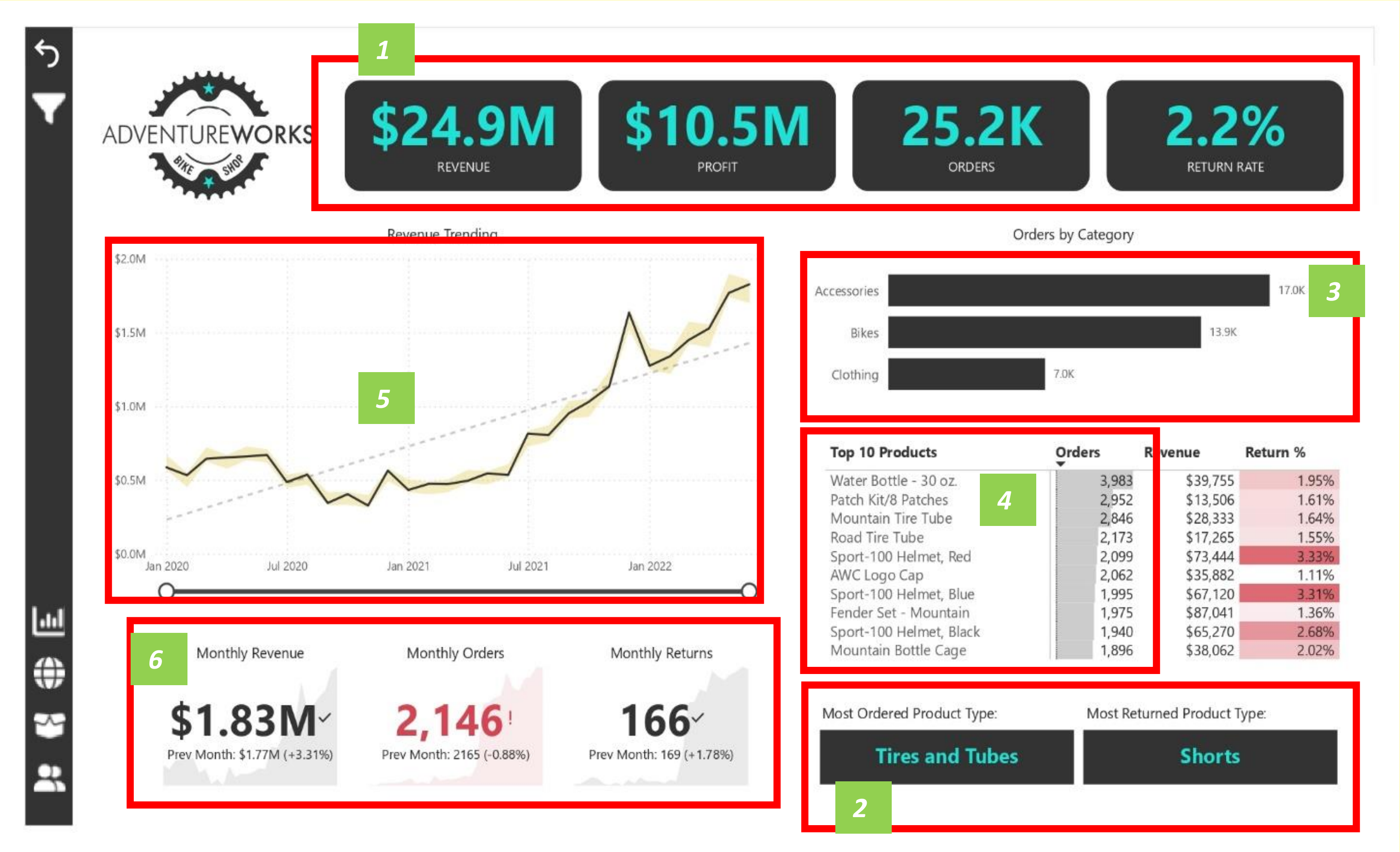
This showcases a sample of a dax measure created to calculate a 10 day Rolling revenue



This shows a table created to house all the DAX measures created in the course of this analysis

Now we answer the business questions using Data Visualization (EDA)

6. Data Visualization



1. What is the business perfoemance based on the KPI (Revenue, Profit, Orders, Return Rate)?
2. What products are the most sold and the most returned? And why?
3. What category of products has the most orders?
4. What are the top 10 products by orders?
5. What is the revenue trend over the years?
6. What is the revenue, profit and returns performances compared with previous months

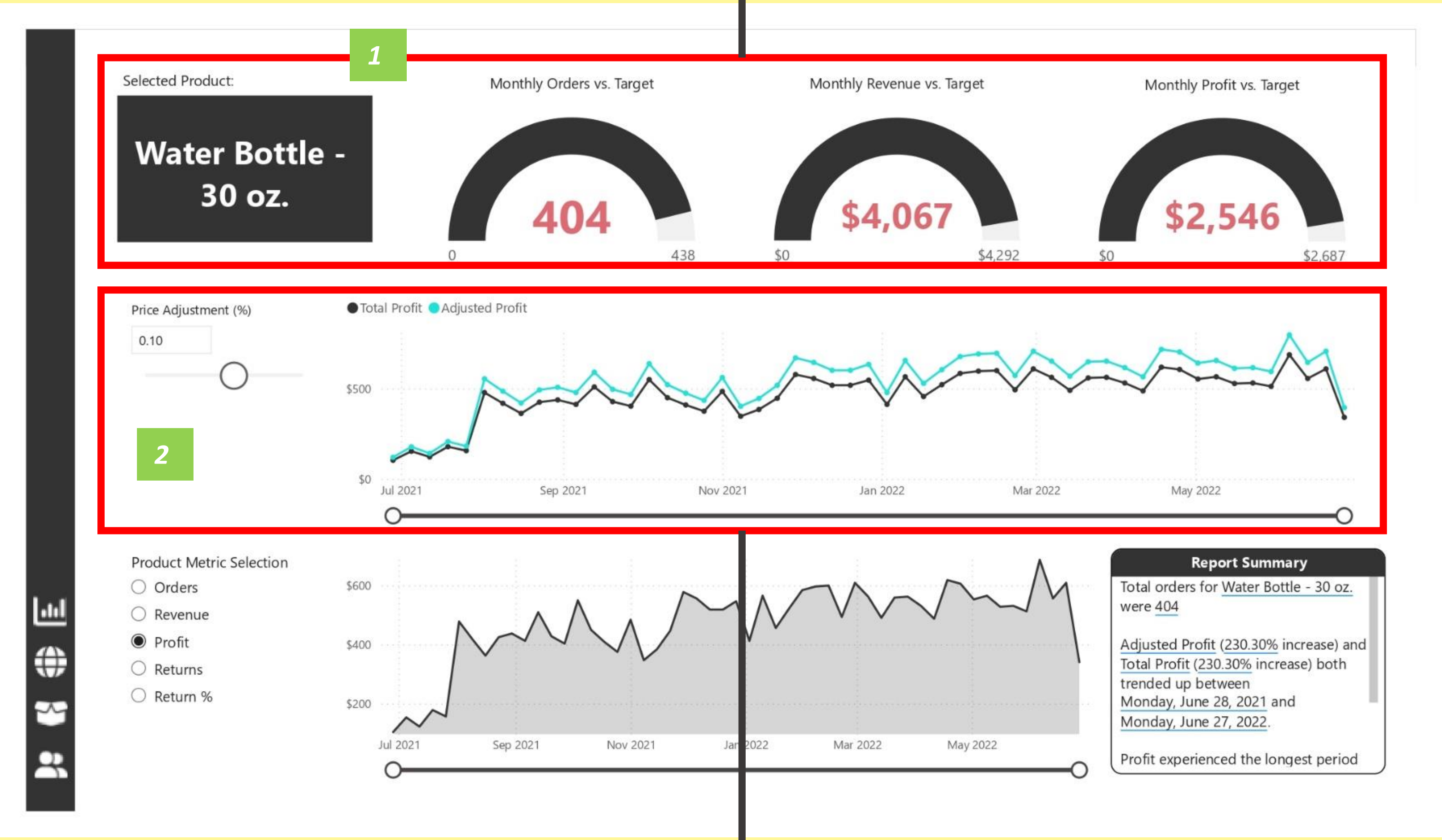
Data Visualization

The bubble size represents order quantiry

1. Which region has produced the most revenue?

Data Visualization

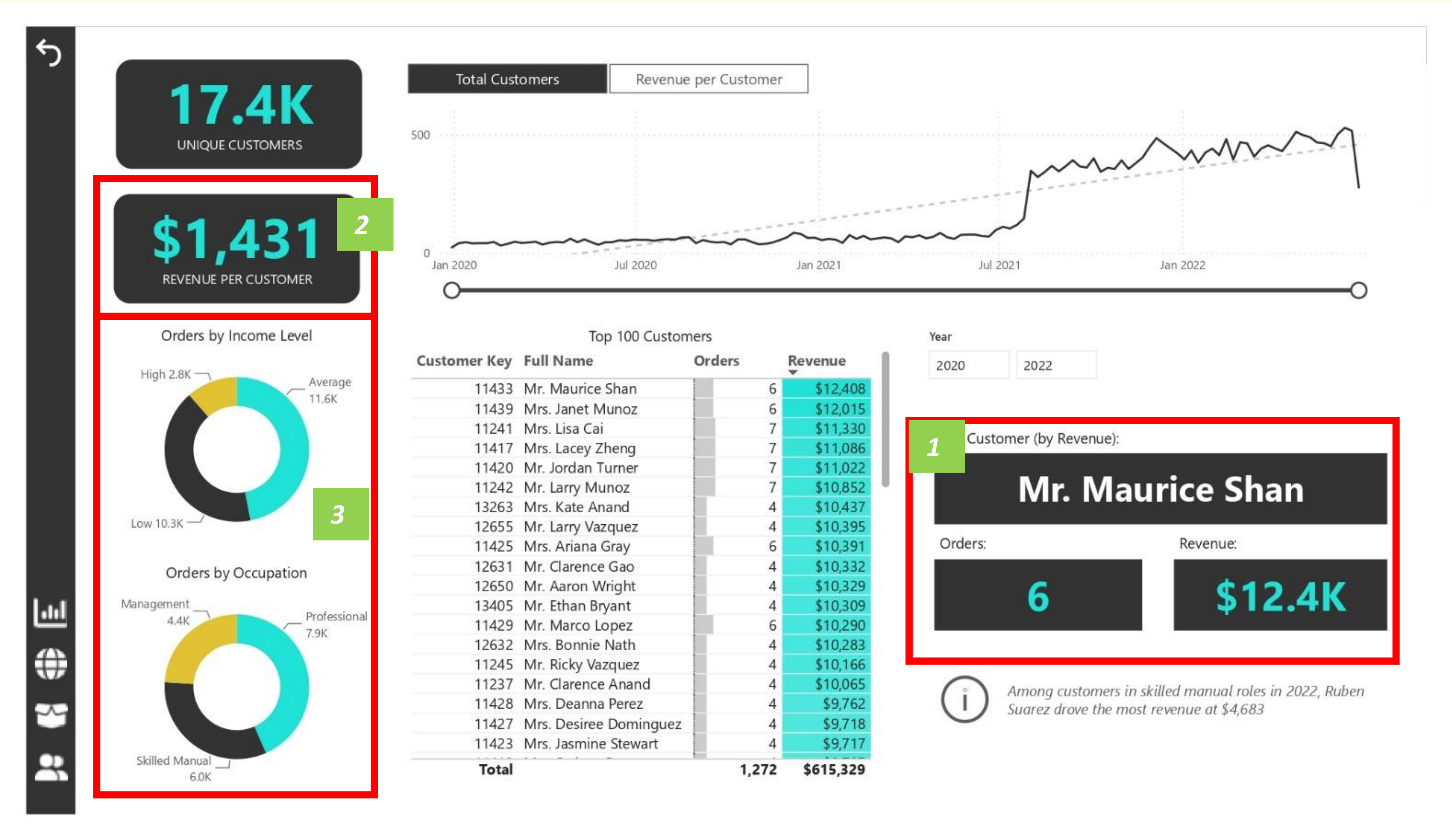
This shows the comparison of orders, profits and revenue between currently monthly amount and target amounts of each product.



1. Predict revenue with a 10% increase in price.
2. How are each product performing?

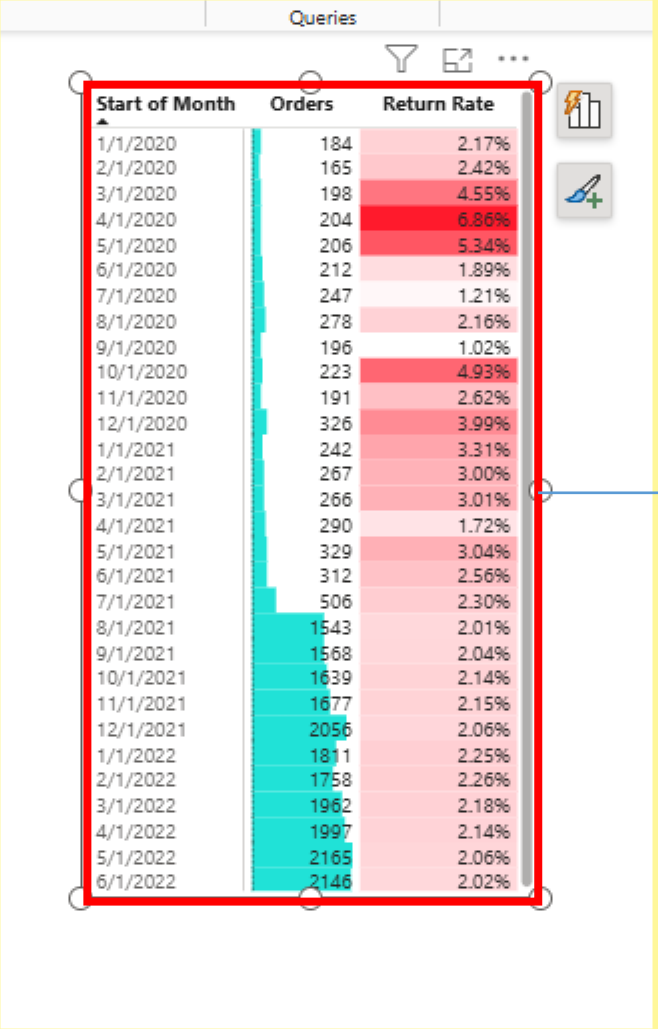
Slider on the far left allows us to view a predicted profit mount if prices were adjusted by a certain percent

Data Visualization



1. Which Customer has invesetd the most over the years?
2. What is the average revenue generated per customer?
3. What is the total order by the top 3 Income level and occupation?

Data Visualization



Queries

Start of Month	Orders	Return Rate
1/1/2020	184	2.17%
2/1/2020	165	2.42%
3/1/2020	198	4.55%
4/1/2020	204	6.86%
5/1/2020	206	5.34%
6/1/2020	212	1.89%
7/1/2020	247	1.21%
8/1/2020	278	2.16%
9/1/2020	196	1.02%
10/1/2020	223	4.93%
11/1/2020	191	2.62%
12/1/2020	326	3.99%
1/1/2021	242	3.31%
2/1/2021	267	3.00%
3/1/2021	266	3.01%
4/1/2021	290	1.72%
5/1/2021	329	3.04%
6/1/2021	312	2.56%
7/1/2021	506	2.30%
8/1/2021	1543	2.01%
9/1/2021	1568	2.04%
10/1/2021	1639	2.14%
11/1/2021	1677	2.15%
12/1/2021	2056	2.06%
1/1/2022	1811	2.25%
2/1/2022	1758	2.26%
3/1/2022	1962	2.18%
4/1/2022	1997	2.14%
5/1/2022	2165	2.06%
6/1/2022	2146	2.02%

1. Why is their an increase in return rates?

Return rates are relatively same in recent times but orders have skyrocketed

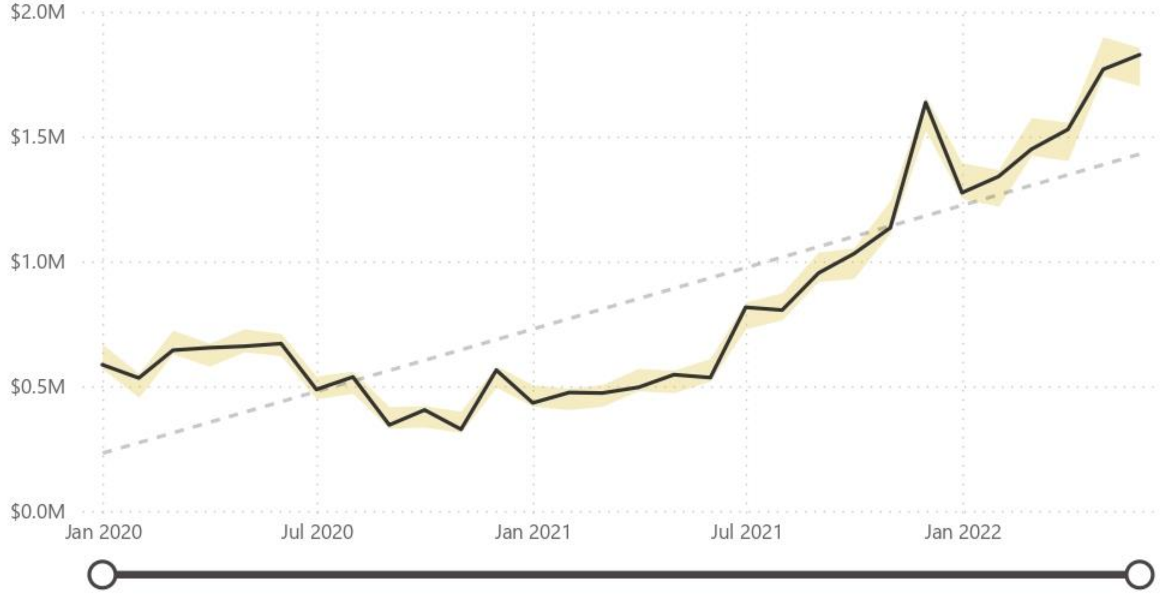
Conclusion

- I've heard people talk about the power of DAX measures, I didn't quite understand why,
- but doing this project, I made use of it 90% of the queries. The range of flexibility it allowed, making possible the exploration of data in ways that calculated columns would struggle at.
- The business didn't have an increaase in return rates as the rates remained relatively constance in recent times but there was a huge increase in orders

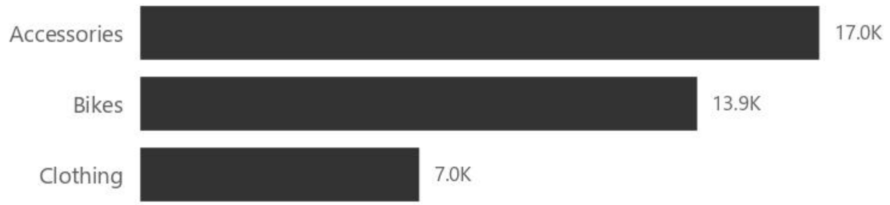
ADVENTUREWORKS BIKE SHOP

$24.9M	$10.5M	25.2K	2.2%
REVENUE	PROFIT	ORDERS	RETURN RATE

Revenue Trending



Orders by Category



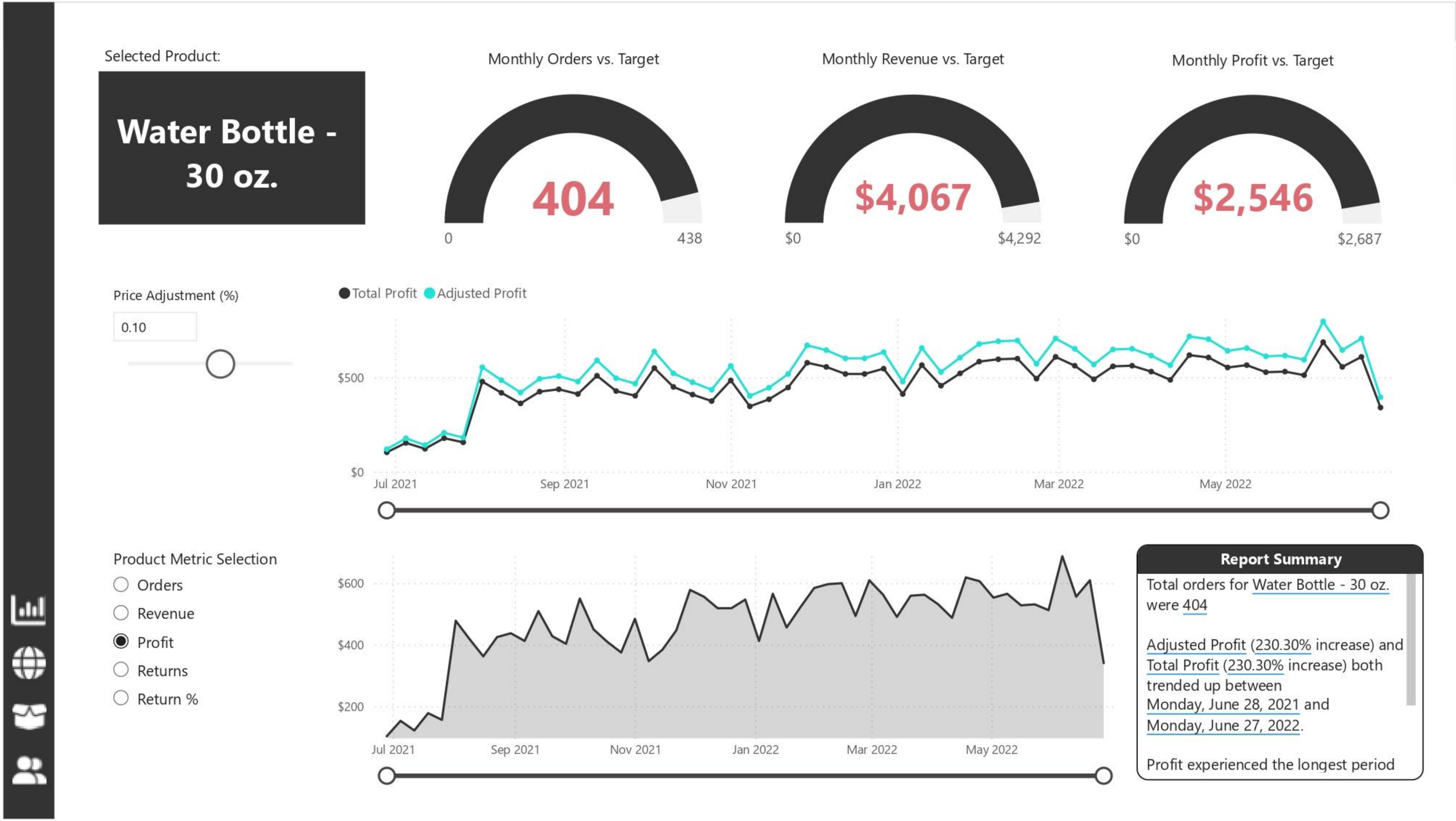
Top 10 Products	Orders	Revenue	Return %
Water Bottle - 30 oz.	3,983	$39,755	1.95%
Patch Kit/8 Patches	2,952	$13,506	1.61%
Mountain Tire Tube	2,846	$28,333	1.64%
Road Tire Tube	2,173	$17,265	1.55%
Sport-100 Helmet, Red	2,099	$73,444	3.33%
AWC Logo Cap	2,062	$35,882	1.11%
Sport-100 Helmet, Blue	1,995	$67,120	3.31%
Fender Set - Mountain	1,975	$87,041	1.36%
Sport-100 Helmet, Black	1,940	$65,270	2.68%
Mountain Bottle Cage	1,896	$38,062	2.02%

Monthly Revenue	Monthly Orders	Monthly Returns
$1.83M ✓	2,146 !	166 ✓
Prev Month: $1.77M (+3.31%)	Prev Month: 2165 (-0.88%)	Prev Month: 169 (+1.78%)

Most Ordered Product Type: **Tires and Tubes**

Most Returned Product Type: **Shorts**

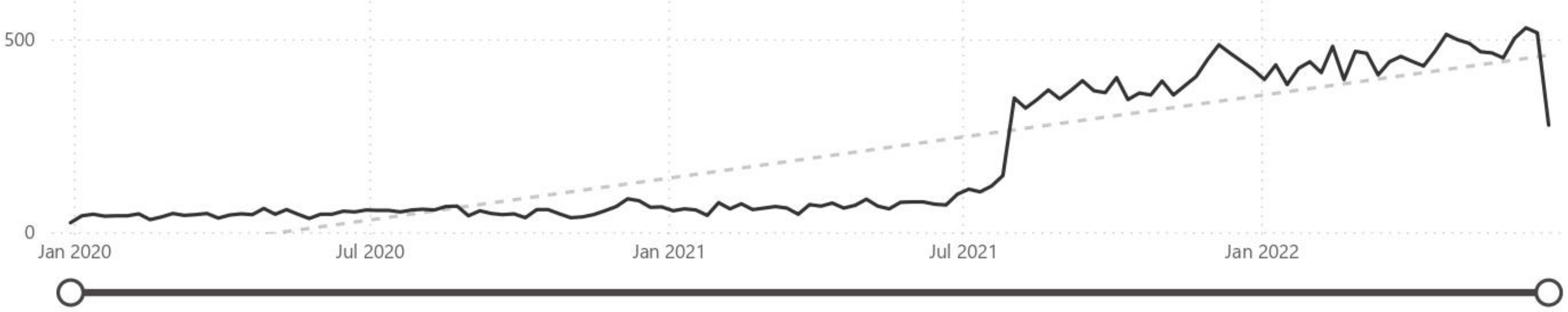

Select all
Europe
North America
Pacific
Canada
NORTH AMERICA
United States
United Kingdom
France
EUROPE
Germany
ASIA
Atlantic Ocean
Pacific Ocean
AFRICA
SOUTH AMERICA
Indian Ocean
Australia
AUSTRALIA
Microsoft Bing


Selected Product:
Water Bottle - 30 oz.
Monthly Orders vs. Target
404
0
438
Monthly Revenue vs. Target
$4,067
$0
$4,292
Monthly Profit vs. Target
$2,546
$0
$2,687
Price Adjustment (%)
0.10
Total Profit
Adjusted Profit
$500
$0
Jul 2021
Sep 2021
Nov 2021
Jan 2022
Mar 2022
May 2022
Product Metric Selection
Orders
Revenue
Profit
Returns
Return %
$600
$400
$200
Jul 2021
Sep 2021
Nov 2021
Jan 2022
Mar 2022
May 2022
Report Summary
Total orders for Water Bottle - 30 oz. were 404
Adjusted Profit (230.30% increase) and Total Profit (230.30% increase) both trended up between Monday, June 28, 2021 and Monday, June 27, 2022.
Profit experienced the longest period

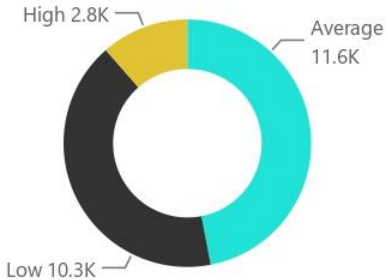
17.4K
UNIQUE CUSTOMERS

$1,431
REVENUE PER CUSTOMER

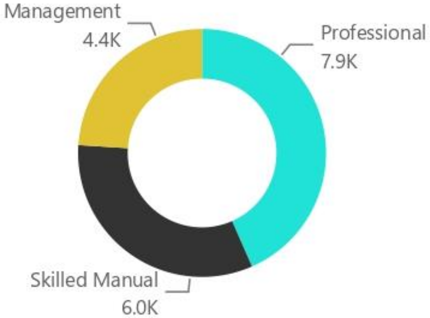
Total Customers | Revenue per Customer



Orders by Income Level



Orders by Occupation



Top 100 Customers

Customer Key	Full Name	Orders	Revenue
11433	Mr. Maurice Shan	6	$12,408
11439	Mrs. Janet Munoz	6	$12,015
11241	Mrs. Lisa Cai	7	$11,330
11417	Mrs. Lacey Zheng	7	$11,086
11420	Mr. Jordan Turner	7	$11,022
11242	Mr. Larry Munoz	7	$10,852
13263	Mrs. Kate Anand	4	$10,437
12655	Mr. Larry Vazquez	4	$10,395
11425	Mrs. Ariana Gray	6	$10,391
12631	Mr. Clarence Gao	4	$10,332
12650	Mr. Aaron Wright	4	$10,329
13405	Mr. Ethan Bryant	4	$10,309
11429	Mr. Marco Lopez	6	$10,290
12632	Mrs. Bonnie Nath	4	$10,283
11245	Mr. Ricky Vazquez	4	$10,166
11237	Mr. Clarence Anand	4	$10,065
11428	Mrs. Deanna Perez	4	$9,762
11427	Mrs. Desiree Dominguez	4	$9,718
11423	Mrs. Jasmine Stewart	4	$9,717
Total		**1,272**	**$615,329**

Year

2020 | 2022

Top Customer (by Revenue):

Mr. Maurice Shan

Orders: **6**

Revenue: **$12.4K**

Among customers in skilled manual roles in 2022, Ruben Suarez drove the most revenue at $4,683